

Joseph M. Ryan

joseph.ryan@duke.edu Languages: English and French (native), Spanish (working)
Background: Winchester College alumnus

Education

Duke University, Ph.D. Candidate in Physics (Quantum Computing) at the Duke Quantum Center. 2022–present
Advisor: Crystal Noel, Theory Collaborators: Stephen Teitsworth.
University of Michigan, B.S. in Honors Physics & Mathematics, High Honors. 2022
Recipient of William L. Williams Award for thesis on QCD color entanglement (LHCb).

Research Summary

Physicist specializing in **quantum stochastic processes**.

Research Experience

Duke Quantum Center (Noel Lab) 2022–present
• Built and operated a trapped-ion quantum computer
Univ. of Michigan – Aidala Group (LHCb Experiment at CERN) 2021–22
• Designed statistical methods to detect TMD factorization breaking due to color entanglement in proton collisions; analyzed correlated stochastic parton dynamics.

Publications

• J. M. Ryan *et al.*, “Experimental Measurement of Quantum First-Passage-Time Distributions,” *under review at PRL* (2025). arXiv:2508.21790. Featured in *Quantum Zeitgeist*.

Selected Talks

• APS DAMOP (2025) – *Noise-Driven Quantum First-Passage-Time Distributions in Ion Traps*
• NSF Robust Quantum Computing, Princeton (2024) – *Quantum Brownian Motion and First-Passage Times*.
• APS March Meeting (2024) – *Noise-Driven Escape Times in Trapped Ion Systems*.

Work Experience

OTC Fellow, Duke Office of Translation & Commercialization 2025–present
• Supporting Duke startups through market analysis and go-to-market strategy development.
Advanced Physics Laboratory, Univ. of Michigan 2020
• Teaching Assistant. Designed and taught advanced atomic, optical and nuclear physics experiments.
Michigan Hyperloop Team 2018–19
• Business director. Raised and managed funds for a multidisciplinary engineering team for hyperloop.
Airbus 2017
• Metrology Intern. Developed precision measurement protocols to ensure quality and accuracy in wing production.

Skills

• **Stochastic Modeling:** Quantum/classical stochastic processes, first-passage times